

A Comparative Analysis Framework for P2P Energy Trading: SDR-Based Dynamic Pricing with Demand-Side Management in Community Microgrids

Alen Shaju, Aarav Kumar, Anshu Singh, Urishita Gupta, and VSKV Harish

Department of Electrical Engineering
Netaji Subhas University of Technology (NSUT)
New Delhi, India
vskvharish@ieee.org

Abstract—This paper presents a modular comparative analysis framework for evaluating peer-to-peer (P2P) energy trading mechanisms in community microgrids. Four settlement strategies (conventional billing, Bill Sharing, Mid-Market Ratio, and Supply-Demand Ratio with Demand-Side Management) are benchmarked on identical community profiles using standardized metrics: cost savings and a fairness index quantifying consumer-prosumer equity. An iterative distributed algorithm resolves the closed-loop price-demand feedback inherent in SDR-DSM. The framework is validated across community sizes from $N = 7$ to $N = 30$ households, demonstrating that MMR and static SDR consistently penalize prosumers while the integration of demand-side management with SDR pricing achieves the highest community savings (up to 18.55%) and fairness simultaneously, delivering positive outcomes for both consumers and prosumers across all configurations tested.

Index Terms—Peer-to-peer energy trading, microgrid pricing, supply-demand ratio, demand-side management, community energy markets

I. INTRODUCTION

Global renewable generation capacity reached 4,448 GW in 2024, with photovoltaic (PV) installations accounting for 75% of new additions [1]. As rooftop PV proliferates, community microgrids have emerged as a localized solution for coordinating distributed energy resources (DERs) through decentralized Peer-to-Peer (P2P) energy sharing and trading [2]. P2P trading creates an internal electricity market where prosumers and consumers exchange energy at prices determined by aggregated community supply and demand, with continuous grid interconnection ensuring reliability [3]. The fundamental concept is illustrated in Fig. 1.

A. Limitations of Existing Approaches

P2P systems can be classified along two dimensions: sharing architecture (centralized, decentralized, distributed) and market design (cost-sharing, auction-based, ratio-based deterministic pricing) [4].

Cost-sharing mechanisms such as the Bill Sharing Mechanism (BSM) pool community resources and redistribute costs based on allocation rules [5]. While simple to implement, BSM



Fig. 1. Concept of P2P energy sharing and trading.

produces static ex-post prices that lack temporal granularity and fail to incentivize real-time load shifting [6].

Auction-based mechanisms require peers to submit bids, after which a market-clearing price is determined [7], [8]. These approaches impose significant communication overhead and demand strategic bidding capability from participants [9]. Game-theoretic formulations [10], [11] share similar complexity concerns for practical deployment.

Ratio-based deterministic pricing, including the Mid-Market Ratio (MMR), derives internal prices directly from

aggregate power quantities without strategic bidding [12], [13]. However, MMR and similar approaches [14] often redistribute costs in ways that penalize high-yield prosumers. Quantitative analysis by Dwivedi et al. [15] demonstrated that conventional P2P mechanisms can reduce prosumer revenues by 8–15% compared to direct grid export, discouraging sustained market participation.

Table I summarizes significant recent works. Existing literature typically addresses pricing, DSM, uncertainty, or network constraints in isolation. A unified dynamic SDR-based pricing mechanism that simultaneously integrates demand flexibility within a scalable microgrid context remains underdeveloped. Recent works have begun integrating DSM with P2P frameworks [16], but a mechanism that dynamically couples supply-demand indexed pricing with demand-responsive load shifting while guaranteeing bounded prices and equitable outcomes is lacking.

B. Contributions

While SDR-based pricing was introduced in [16], the present work makes the following distinct contributions:

- 1) A *modular comparative analysis framework* that evaluates multiple P2P settlement mechanisms (Conventional, BSM, MMR, SDR, SDR-DSM) on identical community profiles using standardized metrics (cost savings and a fairness index quantifying consumer-vs-prosumer equity), enabling systematic benchmarking under controlled conditions.
- 2) Integration of price-responsive DSM with SDR pricing through a distributed iterative algorithm, resolving the closed-loop price–demand feedback that static SDR formulations do not address.
- 3) An *equity-focused evaluation* across community sizes ($N = 7$ to $N = 30$) demonstrating that MMR and static SDR consistently penalize prosumers, while the integration of demand-side management significantly improves both community savings and fairness simultaneously.

II. SYSTEM MODEL

A. Community Microgrid Architecture

The framework is parameterized for an arbitrary number of households N with configurable prosumer-consumer composition and PV sizing categories. Consider a community microgrid with N households comprising PV prosumers $\mathcal{P}$ and consumers $\mathcal{C}$, where $\mathcal{N} = \mathcal{P} \cup \mathcal{C}$. An Energy Service Provider (ESP) acts as the central coordinator for market clearing and settlement. The case study presented herein uses $N = 10$; the formulation generalizes to larger communities without structural modification. The simulation horizon spans H discrete time slots of duration Δt hours.

For household $n \in \mathcal{N}$ at time slot t , the net power is:

$$P_{n,t}^{\text{net}} = D_{n,t} - G_{n,t} \quad (1)$$

where $D_{n,t}$ is the electricity demand and $G_{n,t}$ is PV generation (with $G_{n,t} \equiv 0$ for consumers $n \in \mathcal{C}$). Self-consumption, import, and export powers are:

$$P_{n,t}^{\text{self}} = \min(D_{n,t}, G_{n,t}) \quad (2)$$

$$P_{n,t}^{\text{im}} = \max(P_{n,t}^{\text{net}}, 0), \quad P_{n,t}^{\text{ex}} = \max(-P_{n,t}^{\text{net}}, 0) \quad (3)$$

The grid buying tariff π^{buy} and selling tariff π^{sell} set the upper and lower bounds for internal P2P prices. Community-level reference tariffs are:

$$\bar{\pi}^{\text{buy}} = \frac{1}{N} \sum_{n=1}^N \pi_n^{\text{buy}}, \quad \bar{\pi}^{\text{sell}} = \frac{1}{|\mathcal{P}|} \sum_{n \in \mathcal{P}} \pi_n^{\text{sell}} \quad (4)$$

III. P2P PRICING MECHANISMS

A. Bill Sharing Mechanism (BSM)

The BSM pools community resources over the billing period and computes constant ex-post prices. Total community import and export energies are:

$$E^{\text{buy}} = \sum_t P_t^{\text{buy}} \cdot \Delta t, \quad E^{\text{sell}} = \sum_t P_t^{\text{sell}} \cdot \Delta t \quad (5)$$

where $P_t^{\text{buy}} = \sum_n P_{n,t}^{\text{im}}$ and $P_t^{\text{sell}} = \sum_n P_{n,t}^{\text{ex}}$. The community's net grid interaction determines internal prices:

$$\pi_t^{\text{int,buy}} = \frac{\bar{\pi}^{\text{buy}} \cdot E_{\text{grid}}^{\text{im}}}{E^{\text{buy}}}, \quad \pi_t^{\text{int,sell}} = \frac{\bar{\pi}^{\text{sell}} \cdot E_{\text{grid}}^{\text{ex}}}{E^{\text{sell}}} \quad (6)$$

BSM produces temporally invariant prices, which while ensuring simplicity, provides no incentive for real-time load adjustment.

B. Mid-Market Ratio (MMR) Pricing

The MMR establishes a time-varying midpoint price anchored between grid tariffs:

$$\pi^{\text{mid}} = \frac{\bar{\pi}^{\text{buy}} + \bar{\pi}^{\text{sell}}}{2} \quad (7)$$

At each time slot, the community net balance index is:

$$\eta_t = \frac{P_t^{\text{buy}} - P_t^{\text{sell}}}{P_t^{\text{buy}} + P_t^{\text{sell}}} \quad (8)$$

The P2P clearing price adjusts dynamically:

$$\pi_t^{\text{p2p}} = \pi^{\text{mid}} + \frac{\bar{\pi}^{\text{buy}} - \bar{\pi}^{\text{sell}}}{2} \cdot \eta_t \quad (9)$$

When community supply exceeds demand, the effective selling price is reduced proportionally as surplus energy is exported to the grid at the feed-in tariff. Conversely, during deficit periods, the buying price increases toward the retail tariff. While MMR introduces temporal price variation, it can significantly depress prosumer revenues during surplus periods.

TABLE I
SUMMARY OF SIGNIFICANT WORKS ON P2P ENERGY TRADING ARCHITECTURES

Source	Market Model	Pricing Mechanism	DSM	Optimization	Uncertainty	Test System
Liu (2017)	Centralized P2P	SDR-based pricing	Yes	Iterative optim.	Deterministic	China Southern Grid
Jia (2023)	Receding-horizon	Interval matching + CVar	Yes	Nash bargaining	Prediction intervals	IEEE 13 & 123 bus
Kanakadhurga (2024)	Residential μ grid	SDR + RTP + Double Auction	Yes	Binary Dragonfly	EV uncertainty	4-home microgrid
Ali (2022)	Decentralized P2P	Double auction	Limited	Game-theoretic	Forecast-based	Community microgrid
Veeraswamy (2024)	Distributed P2P	Mid-Market Rate	No	Auction clearing	None	Residential community

C. Supply-Demand Ratio (SDR) Based Pricing

The SDR captures the instantaneous balance between local supply and demand:

$$\text{SDR}_t = \frac{P_t^{\text{sell}}}{P_t^{\text{buy}} + \varepsilon} \quad (10)$$

where ε is a small constant for numerical stability. The SDR maps directly to market regimes: $\text{SDR}_t < 1$ indicates energy shortage, $\text{SDR}_t = 1$ represents balance, and $\text{SDR}_t > 1$ signals surplus.

Internal P2P prices are deterministic functions of the SDR bounded by grid tariffs. The selling price is:

$$\pi_t^{\text{int,sell}} = \frac{\bar{\pi}^{\text{buy}} \cdot \bar{\pi}^{\text{sell}}}{(\bar{\pi}^{\text{buy}} - \bar{\pi}^{\text{sell}}) \cdot \text{SDR}_t + \bar{\pi}^{\text{sell}}} \quad (11)$$

The buying price is defined piecewise:

$$\pi_t^{\text{int,buy}} = \begin{cases} \pi_t^{\text{int,sell}} \cdot \text{SDR}_t + \bar{\pi}^{\text{buy}} \cdot (1 - \text{SDR}_t), & \text{SDR}_t < 1 \\ \bar{\pi}^{\text{sell}}, & \text{SDR}_t \geq 1 \end{cases} \quad (12)$$

Similarly, the selling price (11) is clamped: $\pi_t^{\text{int,sell}} = \bar{\pi}^{\text{sell}}$ when $\text{SDR}_t \geq 1$. This formulation guarantees: (i) price monotonicity with respect to SDR, (ii) bounded prices within $[\bar{\pi}^{\text{sell}}, \bar{\pi}^{\text{buy}}]$, and (iii) zero-arbitrage conditions.

D. SDR with Demand-Side Management

The primary contribution integrates DSM with SDR pricing. Participants adjust consumption $x_{n,t}$ in response to price signals. The net power becomes:

$$P_{n,t}^{\text{net}} = x_{n,t} - G_{n,t} \quad (13)$$

User discomfort from load shifting is modeled as a quadratic penalty [14]:

$$C_n^{\text{rinc}} = \alpha_n \sum_{t=1}^H (x_{n,t} - D_{n,t}^{\text{ref}})^2 \cdot \Delta t \quad (14)$$

where $D_{n,t}^{\text{ref}}$ is the baseline load profile and α_n is the sensitivity coefficient representing tolerance to schedule deviations. The quadratic form is adopted following established practice in demand response literature [14], [16], as it penalizes large deviations disproportionately, reflecting realistic user discomfort where minor adjustments are tolerable but large shifts are increasingly objectionable.

Each participant independently minimizes:

$$\min_{x_{n,t}} \sum_{t=1}^H [\pi_{n,t} \cdot P_{n,t}^{\text{net}} \cdot \Delta t + \alpha_n (x_{n,t} - D_{n,t}^{\text{ref}})^2 \cdot \Delta t] \quad (15)$$

subject to:

$$\sum_{t=1}^H x_{n,t} \cdot \Delta t = \sum_{t=1}^H D_{n,t}^{\text{ref}} \cdot \Delta t \quad (16)$$

$$D_n^{\text{min}} \leq x_{n,t} \leq D_n^{\text{max}}, \quad \forall t \quad (17)$$

Constraint (16) enforces energy conservation, ensuring DSM corresponds strictly to load shifting rather than curtailment. Constraint (17) imposes physical appliance-level bounds reflecting the practical range of deferrable loads (e.g., water heaters, EV charging, washing machines).

The inclusion of DSM introduces a closed-loop interaction: prices depend on SDR, SDR depends on aggregate demand, and demand depends on prices. This feedback loop is resolved through an iterative fixed-point algorithm coordinated by the ESP.

The ESP revenue is derived from a service fee proportional to locally traded energy:

$$S^{\text{ESP}} = \beta \sum_{t=1}^H \min(P_t^{\text{sell}}, P_t^{\text{buy}}) \cdot \Delta t \quad (18)$$

where β covers metering, communication, settlement, and losses.

To quantify equity across participant groups, the framework computes a fairness index:

$$\mathcal{F} = 1 - \frac{|\Delta_C - \Delta_P|}{|\Delta_C| + |\Delta_P| + \varepsilon} \quad (19)$$

where Δ_C and Δ_P are the group-level percentage cost improvements for consumers and prosumers respectively (positive = benefit, negative = loss), relative to conventional billing. $\mathcal{F} = 1$ when both groups benefit equally; $\mathcal{F} \rightarrow 0$ when one group's gain comes entirely at the other's expense.

IV. DISTRIBUTED SETTLEMENT ALGORITHM

The SDR-DSM mechanism is solved through a distributed iterative algorithm. At each iteration, the ESP aggregates prosumer schedules, computes the community SDR, and updates P2P prices. Participants then independently re-optimize their

TABLE II
SIMULATION PARAMETERS

Parameter	Value	Unit
Simulation duration	30	days
Time resolution (Δt)	15	min
Households (N)	10	—
Consumers	3	—
Prosumers (S/M/L)	3/2/2	—
Consumer daily load	4–7	kWh/day
Prosumer daily load	5–9	kWh/day
PV yield	4.5–5.5	kWh/kWp/day
Grid buy tariff (π^{buy})	5.8752	Rs./kWh
Grid sell tariff (π^{sell})	3.2462	Rs./kWh
DSM discomfort coefficient (α)	0.12	Rs. $\cdot$ kW $^{-1}$
DSM damping factor	0.3	—
Max iterations	8	—
ESP service fee (β)	2	%

load schedules given updated prices. The process repeats until convergence.

Algorithm 1: Distributed SDR-DSM Settlement

Input: $D_{n,t}^{\text{ref}}$, $G_{n,t}$, π^{buy} , π^{sell} , α_n , ϵ_{conv}

Output: $x_{n,t}^*$, $\pi_t^{\text{int,buy}}$, $\pi_t^{\text{int,sell}}$

- 1) Initialize $x_{n,t}^{(0)} \leftarrow D_{n,t}^{\text{ref}}$ for all n, t ; set $k \leftarrow 0$
- 2) **Repeat:**
 - (a) ESP aggregates $P_t^{\text{sell},(k)}$, $P_t^{\text{buy},(k)}$ from $\{x_{n,t}^{(k)}\}$
 - (b) Compute $\text{SDR}_t^{(k)}$ via (10)
 - (c) Update prices $\pi_t^{(k)}$ via (11)–(12)
 - (d) Each participant n solves (15) s.t. (16)–(17) to obtain $x_{n,t}^{(k+1)}$
 - (e) $k \leftarrow k + 1$
- 3) **Until** $\max_n \|x_n^{(k)} - x_n^{(k-1)}\| < \epsilon_{\text{conv}}$

Each participant's subproblem is a strictly convex quadratic program with a unique minimizer for given prices. The quadratic penalty α regularizes the demand response: the optimal load adjustment to a price change $\delta\pi$ scales as $\mathcal{O}(\delta\pi/\alpha)$, so larger α dampens per-iteration demand shifts and reduces the feedback gain of the composite mapping. A damping factor of 0.3 is applied to inter-iteration updates to enhance stability. A formal contraction bound is deferred to future work. Empirically, the algorithm is run for 8 iterations with $\alpha = 0.12$ Rs. $\cdot$ kW $^{-1}$; the maximum load change between final iterations falls below 10^{-2} kW across all scenarios tested, indicating practical convergence.

V. SIMULATION SETUP

A. System Configuration

The community microgrid comprises $N = 10$ households: 3 pure consumers and 7 PV prosumers (3 small, 2 medium, 2 large). The simulation runs for 30 days at 15-minute resolution ($\Delta t = 0.25$ h), yielding 96 time slots per day and 2,880 total time steps. Table II summarizes the key parameters.

Demand profiles follow a composite Gaussian shape with morning (08:00) and evening (20:00) peaks, $\pm 5\%$ daily vari-

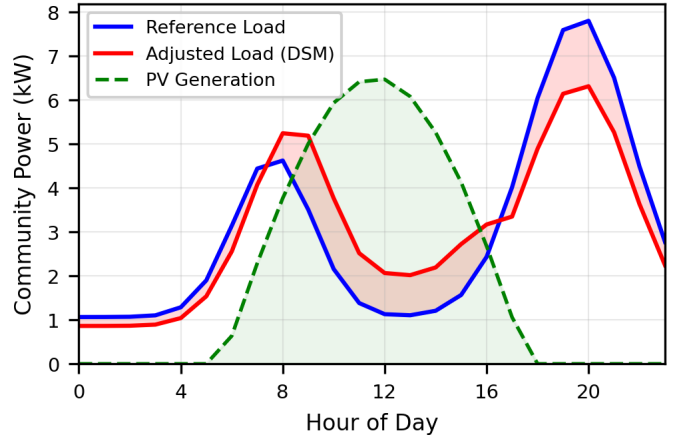


Fig. 2. Aggregate load profile and PV generation for the community microgrid showing reference load, adjusted load, and surplus/deficit regions.

ation, and per-slot noise. PV generation follows a clear-sky sinusoidal curve (06:00–18:00) with $\pm 15\%$ cloud variability. Prosumers are categorized by PV sizing: small (40–60% of load), medium (80–110%), and large (130–180%).

B. Contingency Modeling

Two contingency types are modeled: (i) full-day PV outages due to inverter failure (2 events/month), and (ii) P2P trading suspensions during communication failures (3 events, 2 hours each). Affected peers settle directly with the grid.

C. Software Implementation

The simulation is implemented in Python 3.x using NumPy for numerical computation and Pandas for data management. The iterative SDR-DSM algorithm is implemented as a custom fixed-point solver without external optimization libraries, ensuring computational efficiency and scalability.

VI. RESULTS AND DISCUSSION

A. Community Aggregate Power Profiles

Fig. 2 shows community aggregate power profiles. The adjusted load under SDR-DSM exhibits pronounced midday advancement relative to the reference twin-peak pattern, aligning demand with PV generation. This temporal reallocation mitigates midday exports and enhances local self-consumption.

B. Individual Load Reshaping Under SDR-DSM

Fig. 3 shows load reshaping for representative prosumers. Price signals drive participants to advance deferrable loads into midday intervals (hours 9–15) where high PV surplus produces low clearing prices, while reducing morning and evening consumption. Prosumer 1 (large PV) aggressively shifts demand toward its peak generation period, reducing its evening load substantially. Prosumer 3 (medium PV) exhibits similar but more moderate reshaping, synchronizing load with available solar output. The shaded region illustrates the magnitude of DSM-induced load shifting relative to the reference profile.

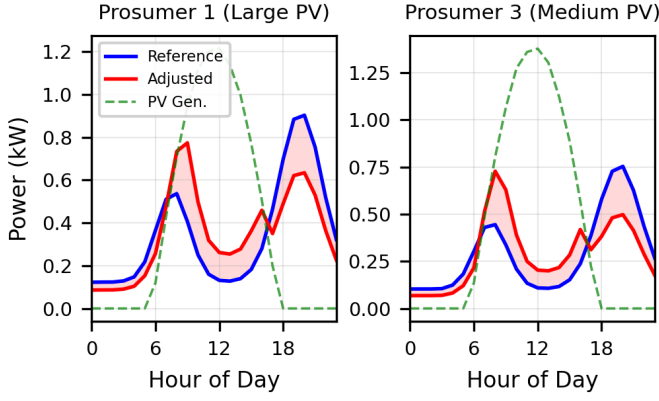


Fig. 3. Sample prosumer load profiles showing reference and adjusted loads under SDR-DSM pricing, with PV generation overlay.

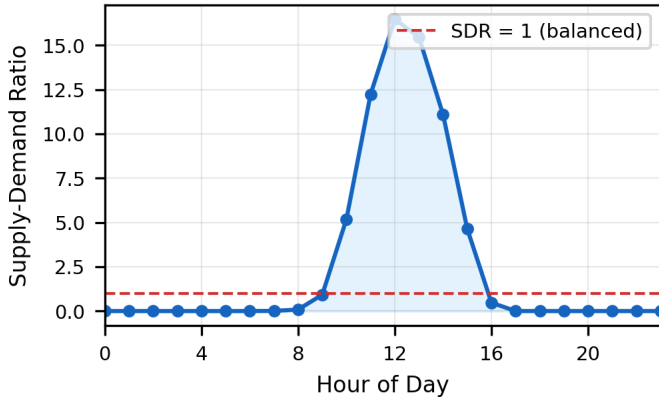


Fig. 4. Hourly Supply-Demand Ratio variation for the community microgrid.

C. SDR Dynamics and Price Formation

The SDR (Fig. 4) partitions the operational horizon into three regimes: morning deficit ($SDR_t \ll 1$), midday surplus ($SDR_t > 1$, peaking at 16.46 during 10:00–14:00), and evening deficit as PV generation ceases.

Fig. 5 shows the corresponding P2P clearing prices. During peak PV periods, internal prices collapse from Rs. 5.8752/kWh to the export floor of Rs. 3.2462/kWh, serving as a natural DSM trigger while remaining strictly bounded between utility limits.

D. Comparative Price Analysis

Fig. 6 compares all four pricing mechanisms. The SDR-based mechanism exhibits the strongest regime-dependent responsiveness, with prices collapsing toward $\bar{\pi}^{\text{sell}}$ during surplus and converging toward $\bar{\pi}^{\text{buy}}$ during deficit. Table III quantifies the differences.

E. Economic Evaluation

Table IV presents comparative economic outcomes. MMR achieves high consumer savings (19.15% group-level) but depreciates prosumer revenues by 5.13%, yielding $\mathcal{F} = 0.00$

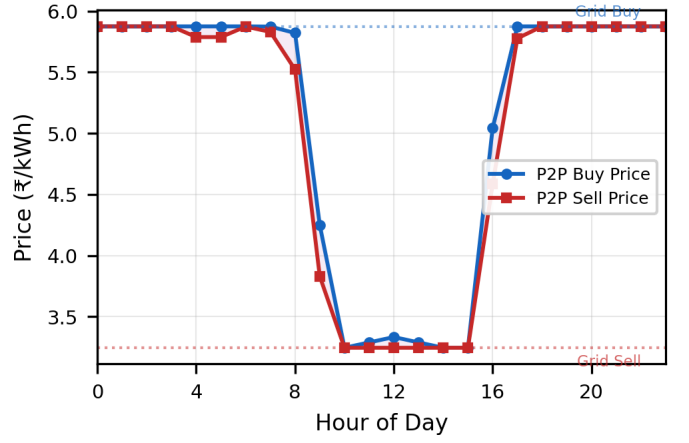


Fig. 5. Hourly internal P2P market clearing prices compared with grid tariffs.

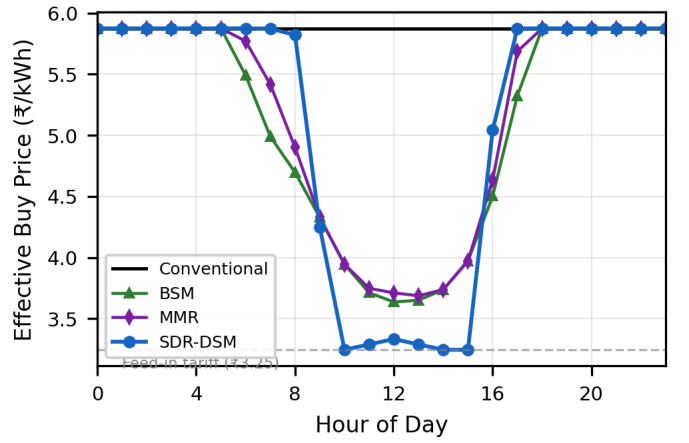


Fig. 6. Comparative validation of SDR-based P2P trading prices with conventional, BSM, and MMR mechanisms.

(maximally inequitable). Static SDR similarly penalizes prosumers (-12.63%) despite strong consumer benefits. BSM achieves moderate fairness ($\mathcal{F} = 0.67$) but with limited savings. The SDR-DSM framework achieves $\mathcal{F} = 0.80$ alongside 15.35% community cost reduction, 18.15% consumer savings, and 11.98% prosumer revenue enhancement.

To validate configurability, Table V presents SDR-DSM results across three additional community sizes ($N = 7$ to $N = 30$). The framework scales without structural modification, consistently achieving the highest fairness and positive savings for both participant groups across all configurations.

VII. CONCLUSION

This paper presented a modular comparative analysis framework for benchmarking P2P energy trading mechanisms under standardized conditions. Systematic evaluation across community sizes from $N = 7$ to $N = 30$ revealed that aggregate cost savings and prosumer equity are inversely coupled in MMR and static SDR pricing. For the primary $N = 10$ community, SDR-DSM achieves 15.35% community savings with $\mathcal{F} = 0.80$, compared to MMR which achieves 8.13%

TABLE III
COMPARATIVE INTRADAY PRICE EXTREMES AND REGIME
CHARACTERISTICS

Metric	Conv.	BSM	MMR	SDR
Max buy (Rs./kWh)	5.88	5.88	5.88	5.88
Min sell (Rs./kWh)	3.25	3.64	3.69	3.25
Avg midday buy (Rs./kWh)	5.88	3.86	3.88	3.41
Price spread (Rs./kWh)	2.63	2.24	2.56	2.63

TABLE IV
30-DAY ECONOMIC PERFORMANCE COMPARISON

Metric	BSM	MMR	SDR	SDR-DSM
Community savings (%)	6.55	8.13	5.32	15.35
Consumer Δ_C (%)	8.47	19.15	20.25	18.15
Prosumer Δ_P (%)	+4.23	-5.13	-12.63	+11.98
Fairness $\mathcal{F}$	0.67	0.00	0.00	0.80

TABLE V
COMMUNITY SIZE SWEEP: SDR-DSM PERFORMANCE

Configuration	Savings (%)	Δ_C (%)	Δ_P (%)	$\mathcal{F}$
2C + 5P ($N = 7$)	18.55	20.35	14.72	0.84
5C + 15P ($N = 20$)	14.93	16.93	13.34	0.88
10C + 20P ($N = 30$)	13.64	15.38	11.18	0.84

savings but at $\mathcal{F} = 0.00$ due to prosumer penalization. The community size sweep confirms scalability, with SDR-DSM consistently achieving positive savings for both participant groups across all configurations tested.

Future work will extend the framework with ACOPF network constraints, stochastic optimization for forecast uncertainty, and BESS/V2G modules for temporal energy arbitrage.

REFERENCES

- [1] IRENA, "Renewable capacity statistics 2025," International Renewable Energy Agency, Abu Dhabi, 2025.
- [2] T. Sousa *et al.*, "Peer-to-peer and community-based markets: A comprehensive review," *Renew. Sustain. Energy Rev.*, vol. 104, pp. 367–378, 2019.
- [3] Y. Zhou and P. D. Lund, "Peer-to-peer energy sharing and trading of renewable energy in smart communities," *Renew. Energy*, vol. 207, pp. 177–193, 2023.
- [4] M. Domènech Monfort *et al.*, "A review of peer-to-peer energy trading with standard terminology proposal," *Energies*, vol. 15, no. 23, p. 9070, 2022.
- [5] C. Long, J. Wu, C. Zhang, M. Cheng, and A. Al-Wakeel, "Peer-to-peer energy trading in a community microgrid," in *Proc. IEEE PES Gen. Meeting*, 2017, pp. 1–5.
- [6] C. Zhang, J. Wu, Y. Zhou, M. Cheng, and C. Long, "Peer-to-peer energy trading in a microgrid," *Appl. Energy*, vol. 220, pp. 1–12, 2018.
- [7] W. Huang *et al.*, "Auction-based P2P energy trading mechanisms," *Appl. Energy*, 2025.
- [8] C. Zhang *et al.*, "Auction-based peer-to-peer energy trading considering echelon utilization of retired EV batteries," *Appl. Energy*, vol. 358, p. 122592, 2024.
- [9] S. Liaquat *et al.*, "Comparative analysis of peer-to-peer PV trading strategies under network constraints," *Appl. Sci.*, vol. 13, no. 18, p. 10044, 2023.
- [10] C. Long, Y. Zhou, and J. Wu, "A game theoretic approach for peer to peer energy trading," *Energy Procedia*, vol. 159, pp. 454–459, 2019.
- [11] W. Tushar *et al.*, "A motivational game-theoretic approach for peer-to-peer energy trading in the smart grid," *Appl. Energy*, vol. 243, pp. 10–20, 2019.
- [12] A. Bogensperger, J. Ferstl, and Y. Yu, "Comparison of pricing mechanisms in peer-to-peer energy communities," in *Proc. Int. Energiewirtschaftstagung*, 2021, p. 12.
- [13] F. Teotia, P. Mathuria, and R. Bhakar, "Peer-to-peer local electricity market platform pricing strategies," *IET Gener. Transm. Distrib.*, vol. 14, no. 20, pp. 4388–4397, 2020.
- [14] F. Alfaverh, M. Denai, and Y. Sun, "A dynamic peer-to-peer electricity market model for a community microgrid with price-based demand response," *IEEE Trans. Smart Grid*, vol. 14, no. 5, pp. 3976–3991, 2023.
- [15] D. Dwivedi *et al.*, "Evaluation of energy resilience and cost benefit in microgrid with peer-to-peer energy trading," *arXiv:2212.02318*, 2022.
- [16] N. Liu *et al.*, "Energy-sharing model with price-based demand response for microgrids of peer-to-peer prosumers," *IEEE Trans. Power Syst.*, vol. 32, no. 5, pp. 3569–3583, 2017.